RAG and AI Quality - Short Study Notes

What is RAG?
Retrieval-Augmented Generation means: find relevant document pieces first, then ask the language model to answer using those pieces.
The goal is fewer hallucinations and answers grounded in your files.

Pipeline stages:
1. Ingest: load PDFs and extract text.
2. Chunk: split text into smaller overlapping windows.
3. Embed: turn each chunk into a numeric vector of meaning.
4. Store: save vectors in a vector database such as Chroma or FAISS.
5. Retrieve: embed the question and find the closest chunks.
6. Generate: LLM answers using retrieved context and should cite sources.

Chunk size tradeoffs:
Chunks that are too small may miss surrounding context.
Chunks that are too large add noise and weaken search precision.
Overlap helps keep sentence context across chunk boundaries.

When the documents do not contain the answer, a good system should say I do not know, instead of inventing facts.

Evaluation ideas for a portfolio RAG app:
Use at least 10 hand-written questions.
Measure retrieval hit-rate at k equals 3: was the right chunk in the top 3?
Record average latency and note known failure modes.

RAG vs fine-tuning:
Use RAG when knowledge lives in changing documents and must be citable.
Use fine-tuning more for style, format, or behavior, not daily PDF updates.